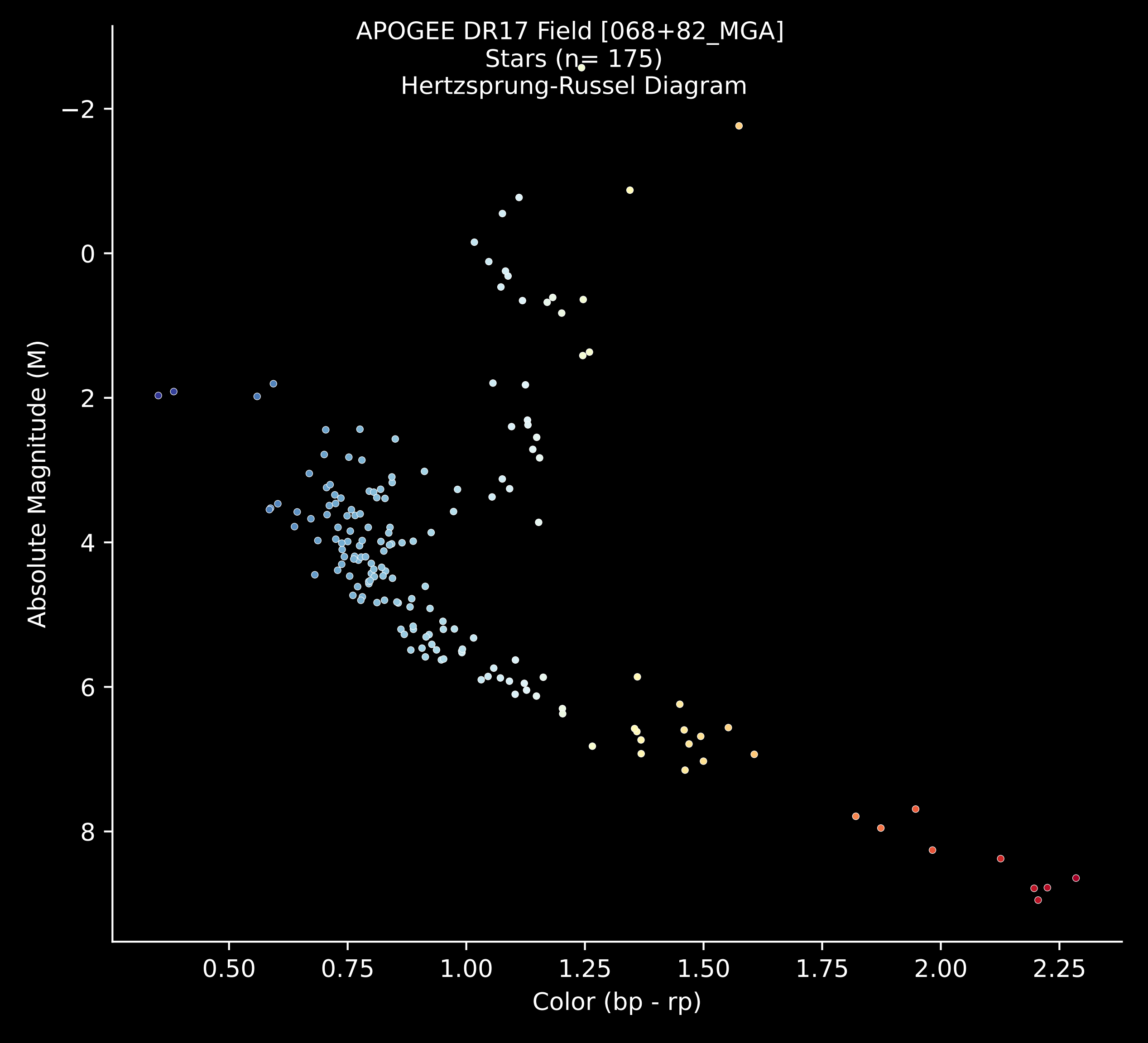
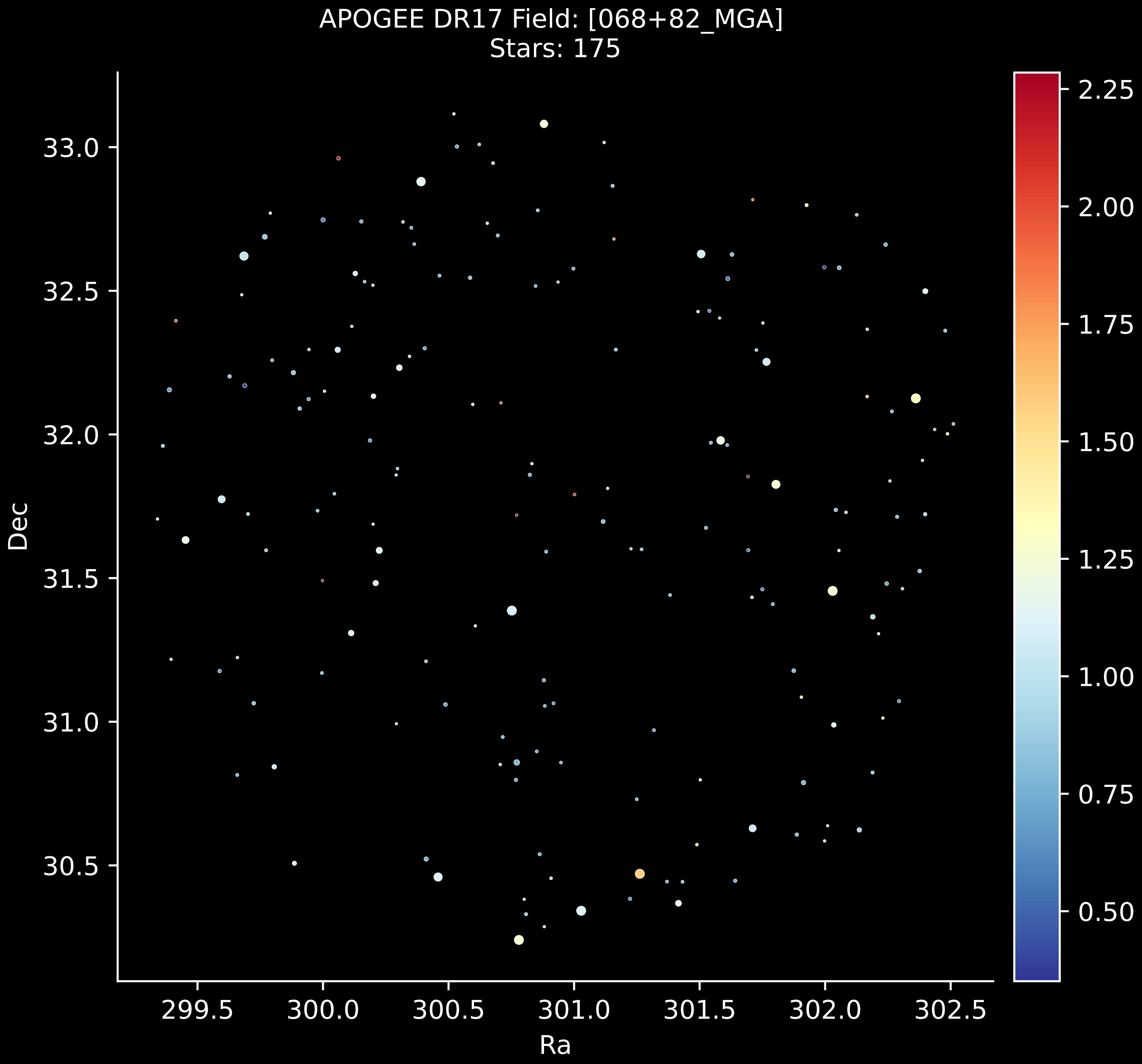
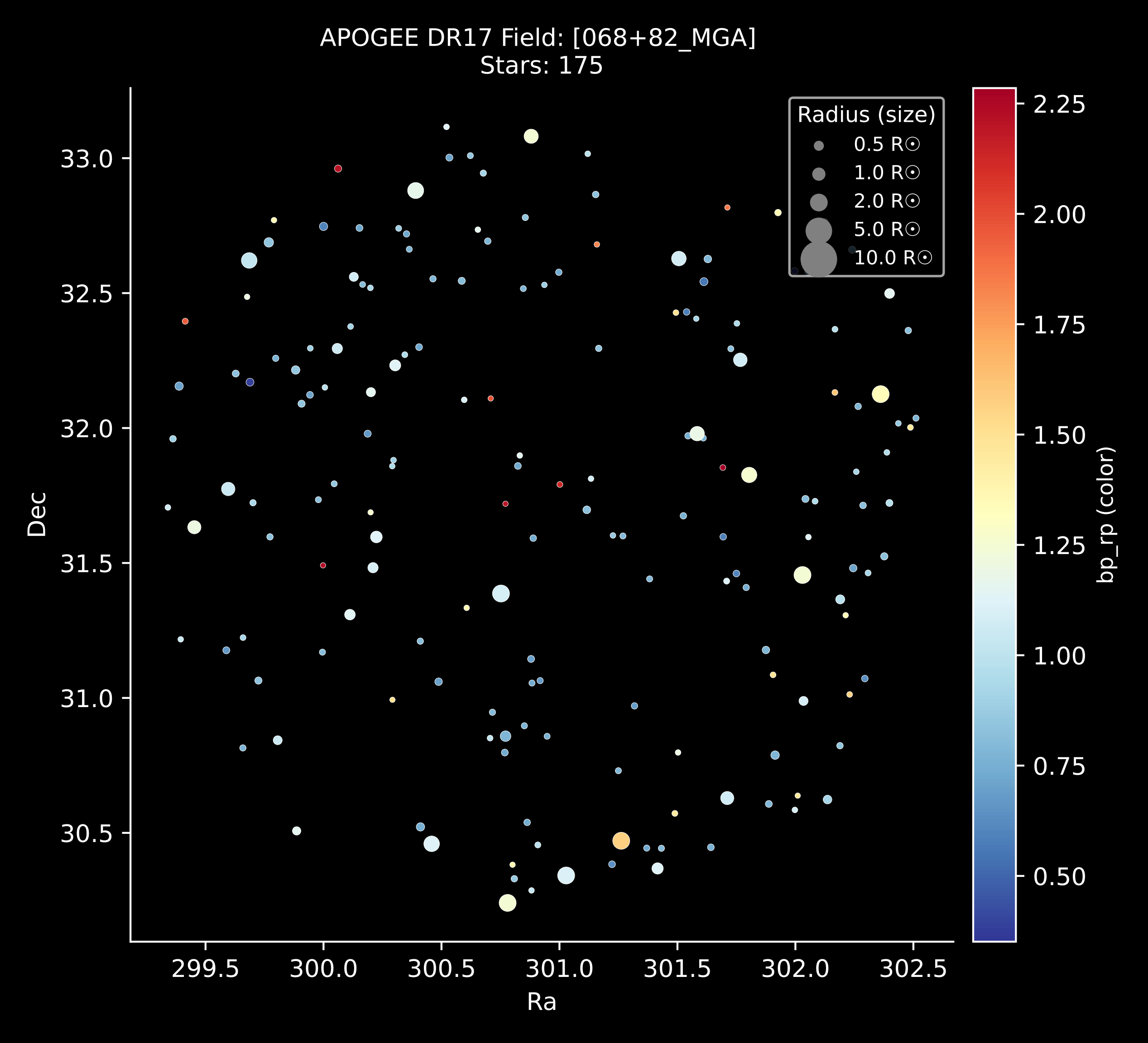
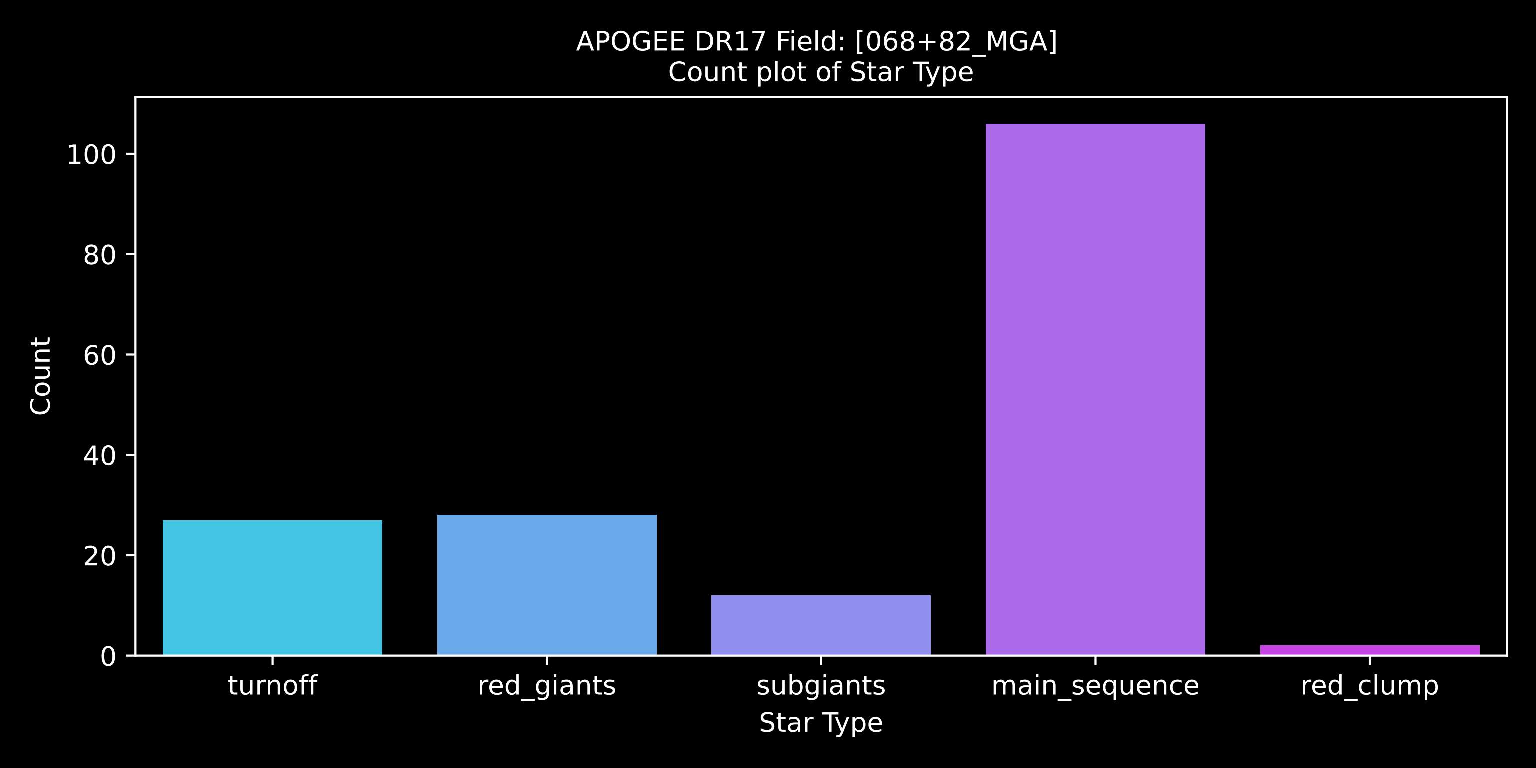
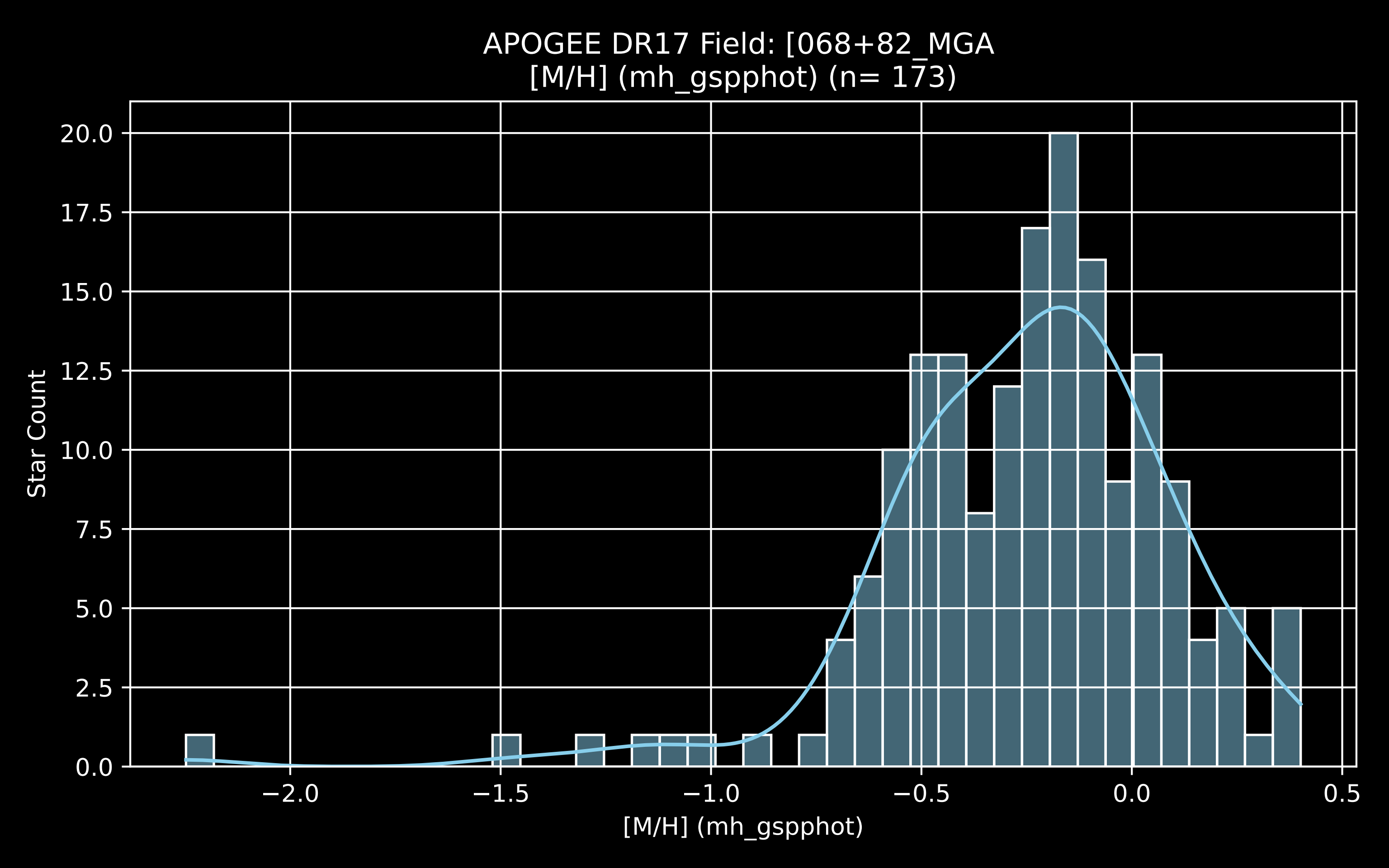

APOGEE DR17 Field [068+82_MGA]
Stars (n= 175)
Hertzsprung-Russel Diagram
Absolute Magnitude (M)
Color (bp - rp)
−2
0
2
4
6
8
0.50
0.75
1.00
1.25
1.50
1.75
2.00
2.25


APOGEE DR17 Field: [068+82_MGA]
Stars: 175
Dec
Ra
bp_rp
33.0
32.5
32.0
31.5
31.0
30.5
299.5
300.0
300.5
301.0
301.5
302.0
302.5
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50

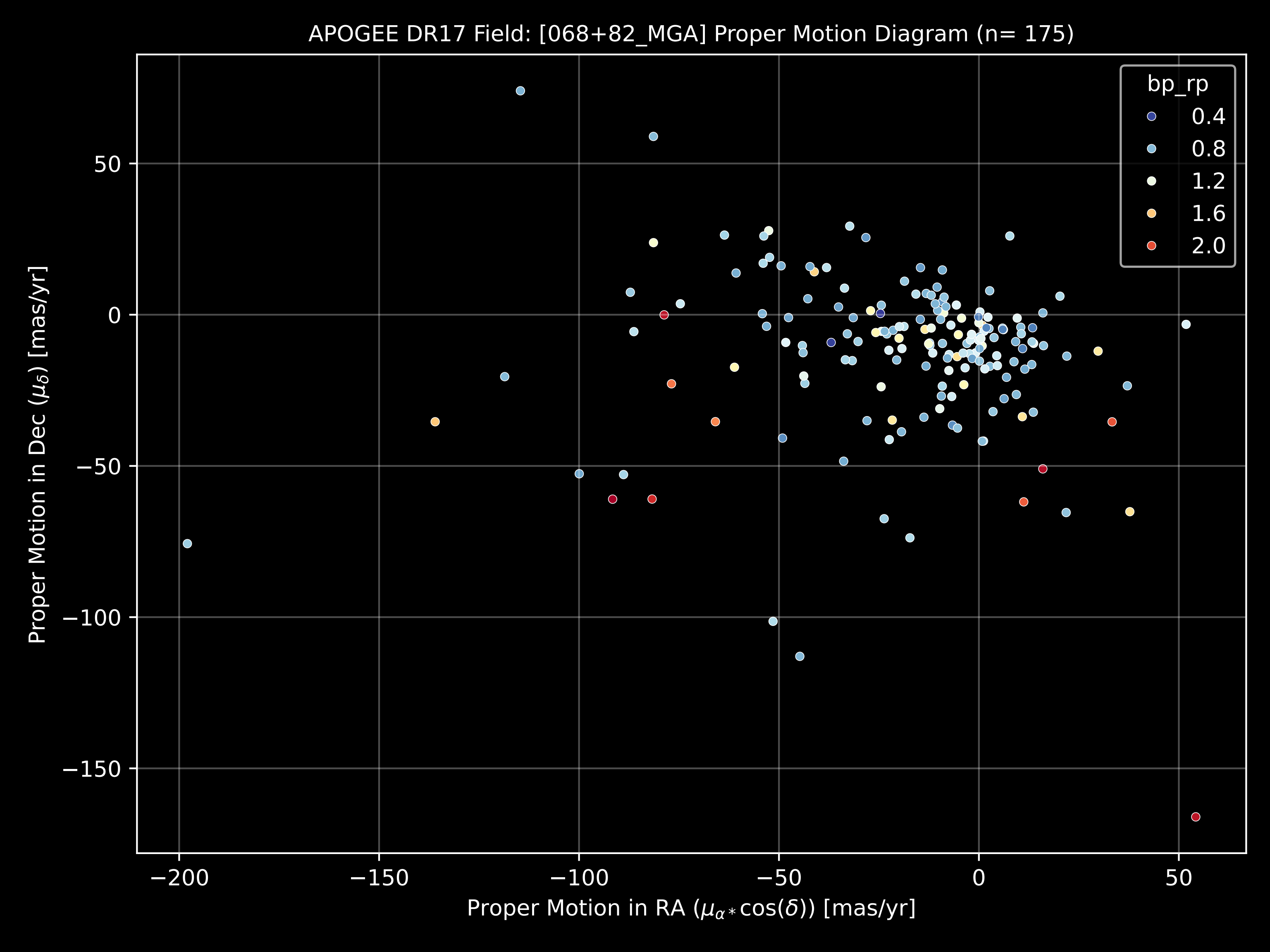


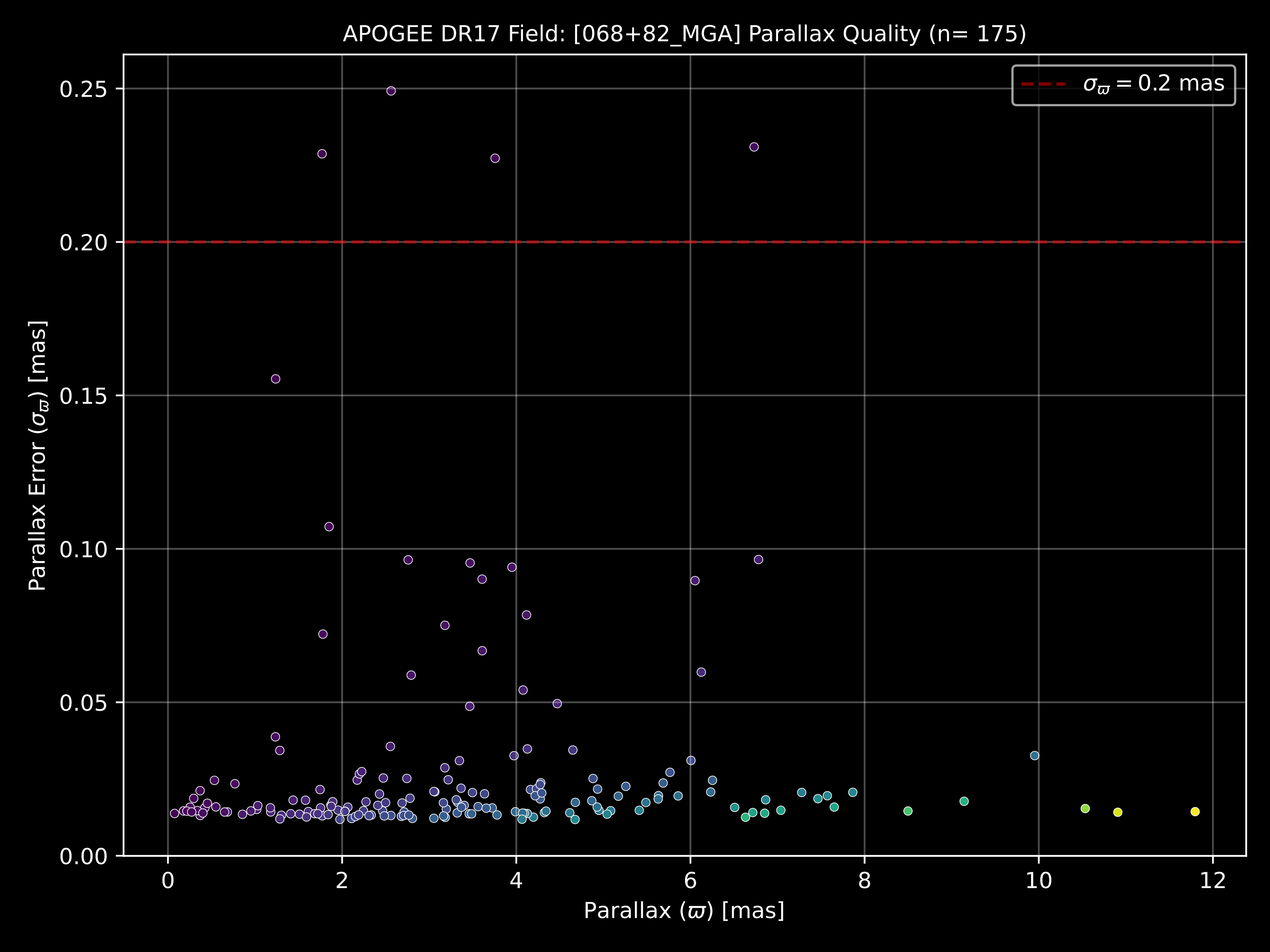
APOGEE DR17 Field: [068+82_MGA]
Stars: 175
Radius (size)
0.5 R☉
1.0 R☉
2.0 R☉
5.0 R☉
10.0 R☉
Dec
Ra
bp_rp (color)
33.0
32.5
32.0
31.5
31.0
30.5
299.5
300.0
300.5
301.0
301.5
302.0
302.5
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50

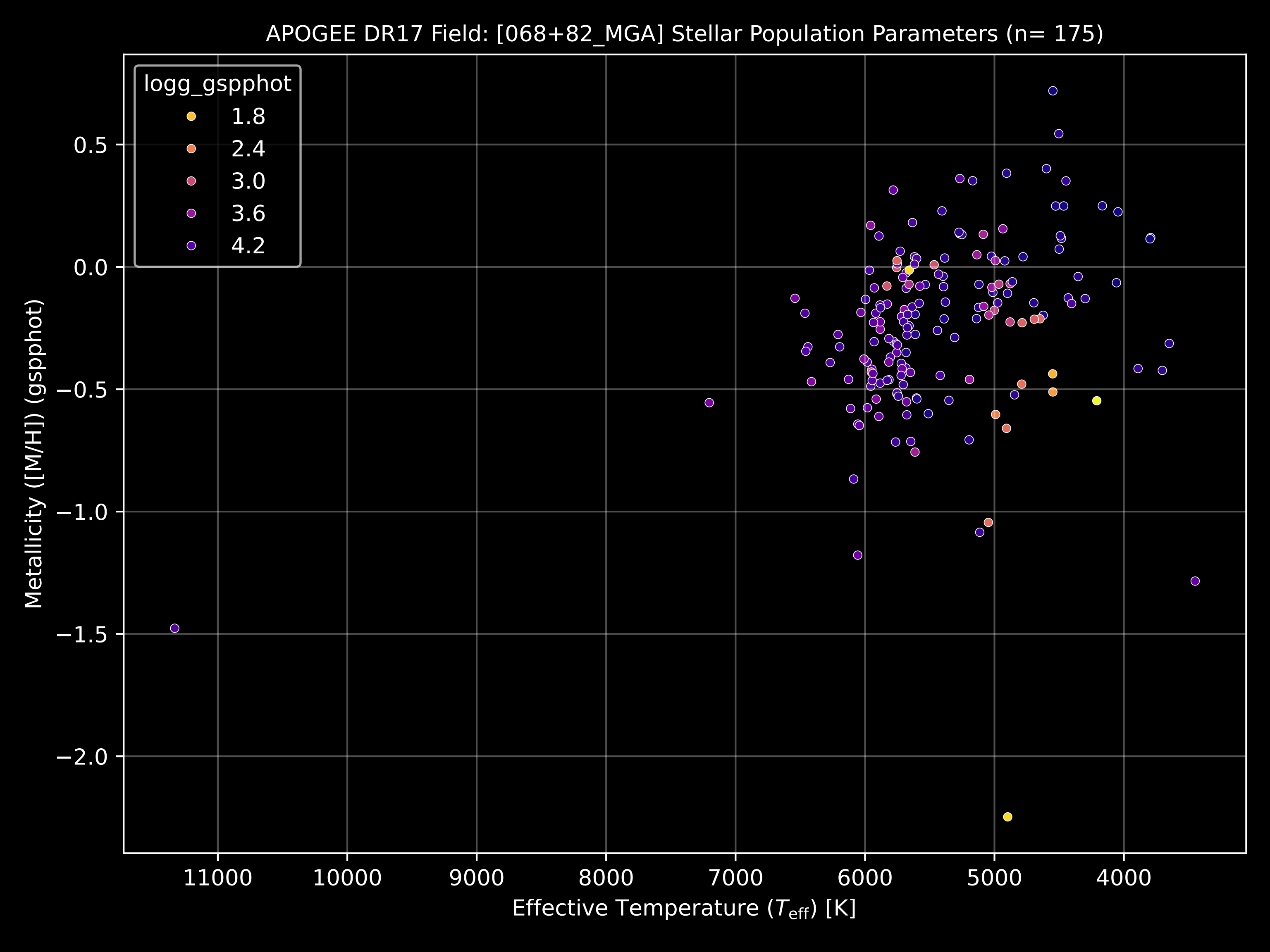

APOGEE DR17 Field: [068+82_MGA]
Count plot of Star Type
Count
0
20
40
60
80
100
turnoff
red_giants
subgiants
main_sequence
red_clump
Star Type



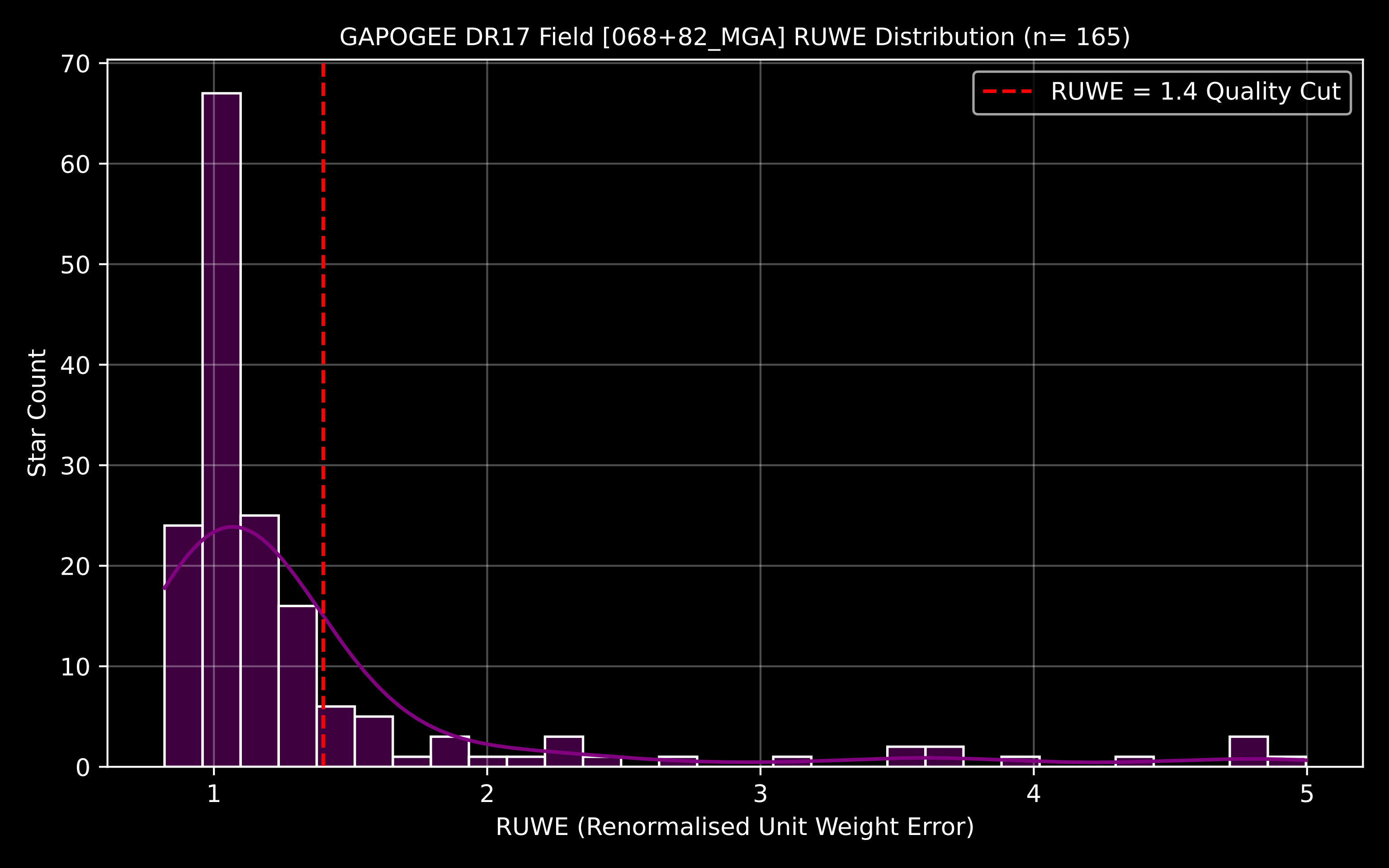
APOGEE DR17 Field: [068+82_MGA
[M/H] (mh_gspphot) (n= 173)
Star Count
[M/H] (mh_gspphot)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
−2.0
−1.5
−1.0
−0.5
0.0
0.5

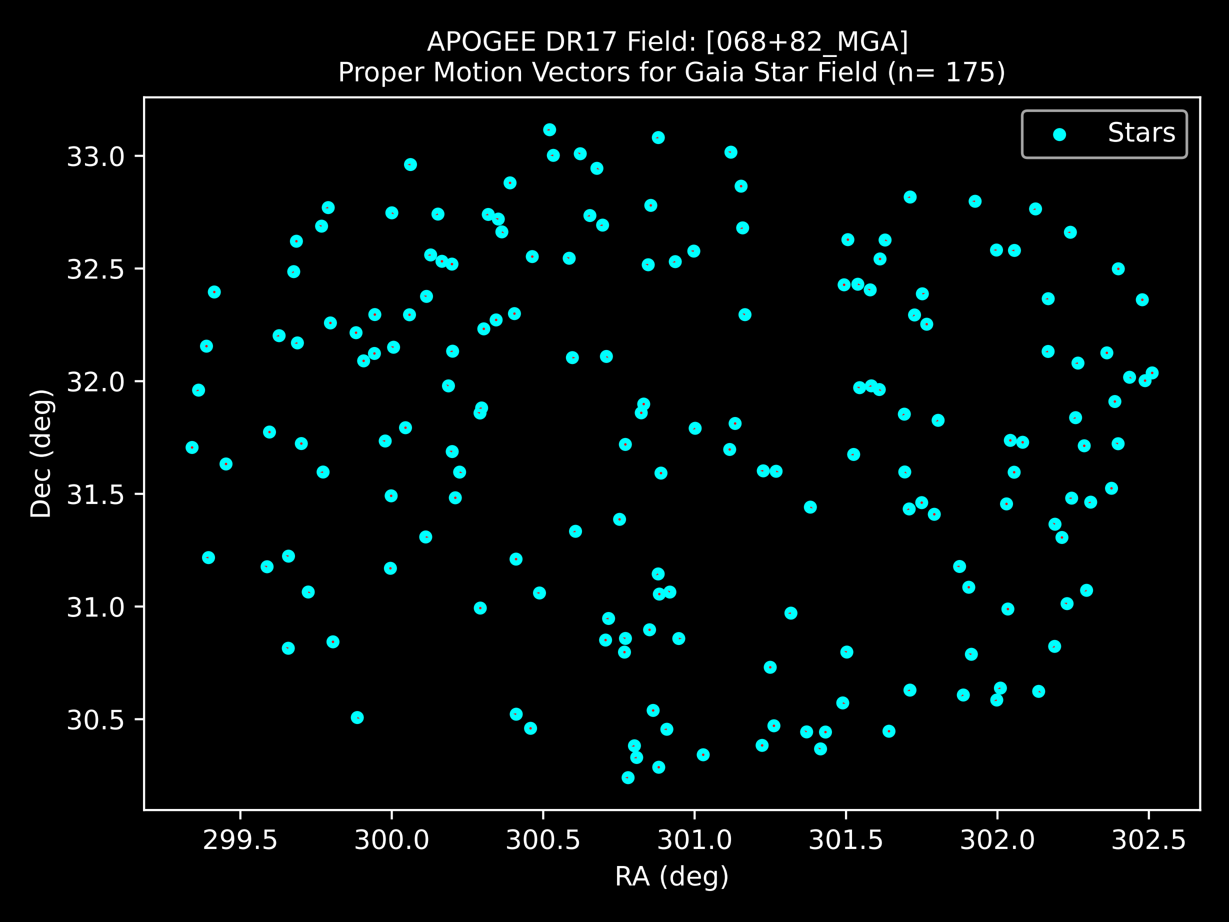

APOGEE DR17 Field: [068+82_MGA] Proper Motion Diagram (n= 175)
Proper Motion in Dec (μ_δ) [mas/yr]
Proper Motion in RA ($\mu_{\alpha*}\cos(\delta)$) [mas/yr]
bp_rp
0.4
0.8
1.2
1.6
2.0
50
0
−50
−100
−150
−200
−150
−100
−50
0
50

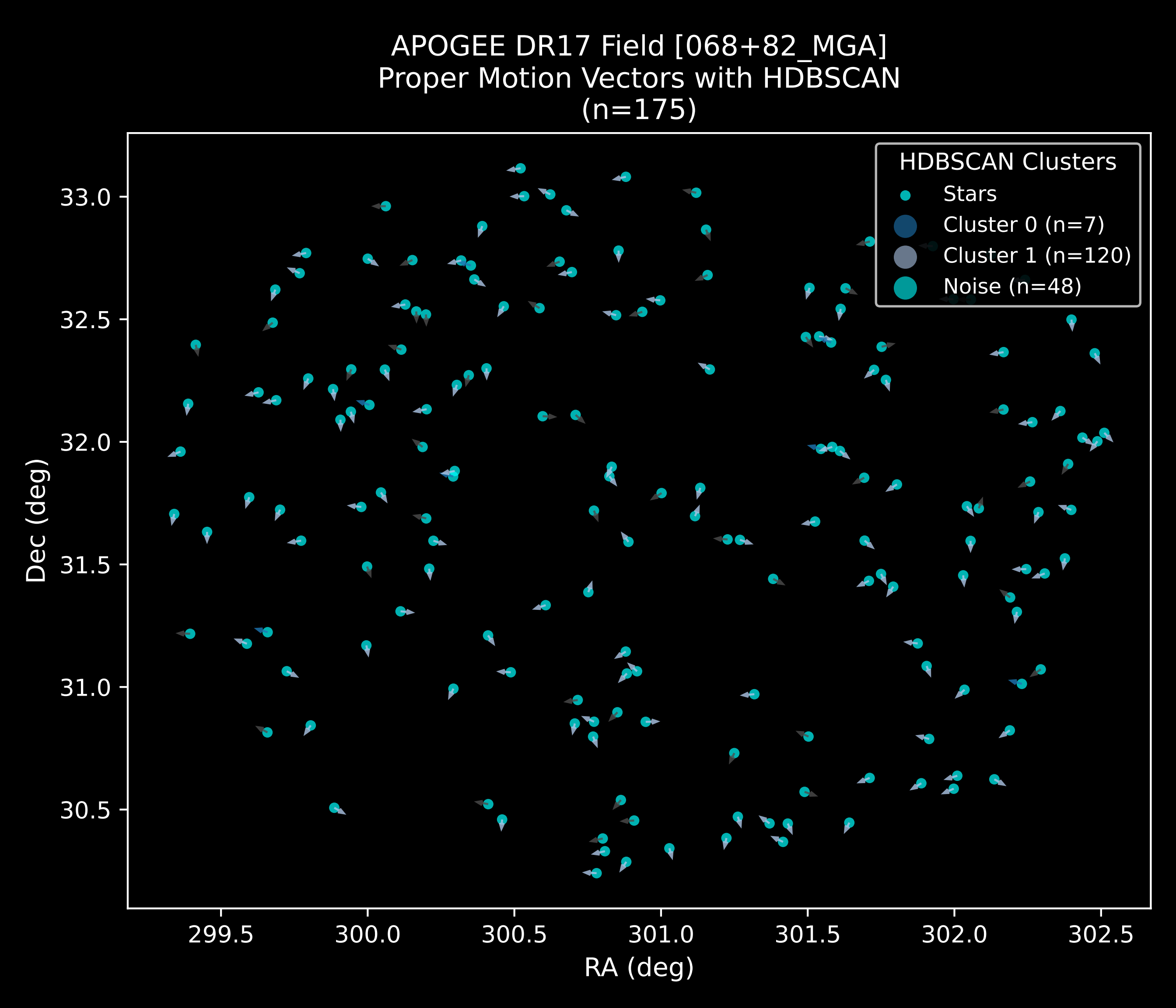

APOGEE DR17 Field: [068+82_MGA] Parallax Quality (n= 175)
$\sigma_\varpi = 0.2$ mas
Parallax Error (σ_ϖ) [mas]
Parallax (ϖ) [mas]
0.00
0.05
0.10
0.15
0.20
0.25
0
2
4
6
8
10
12

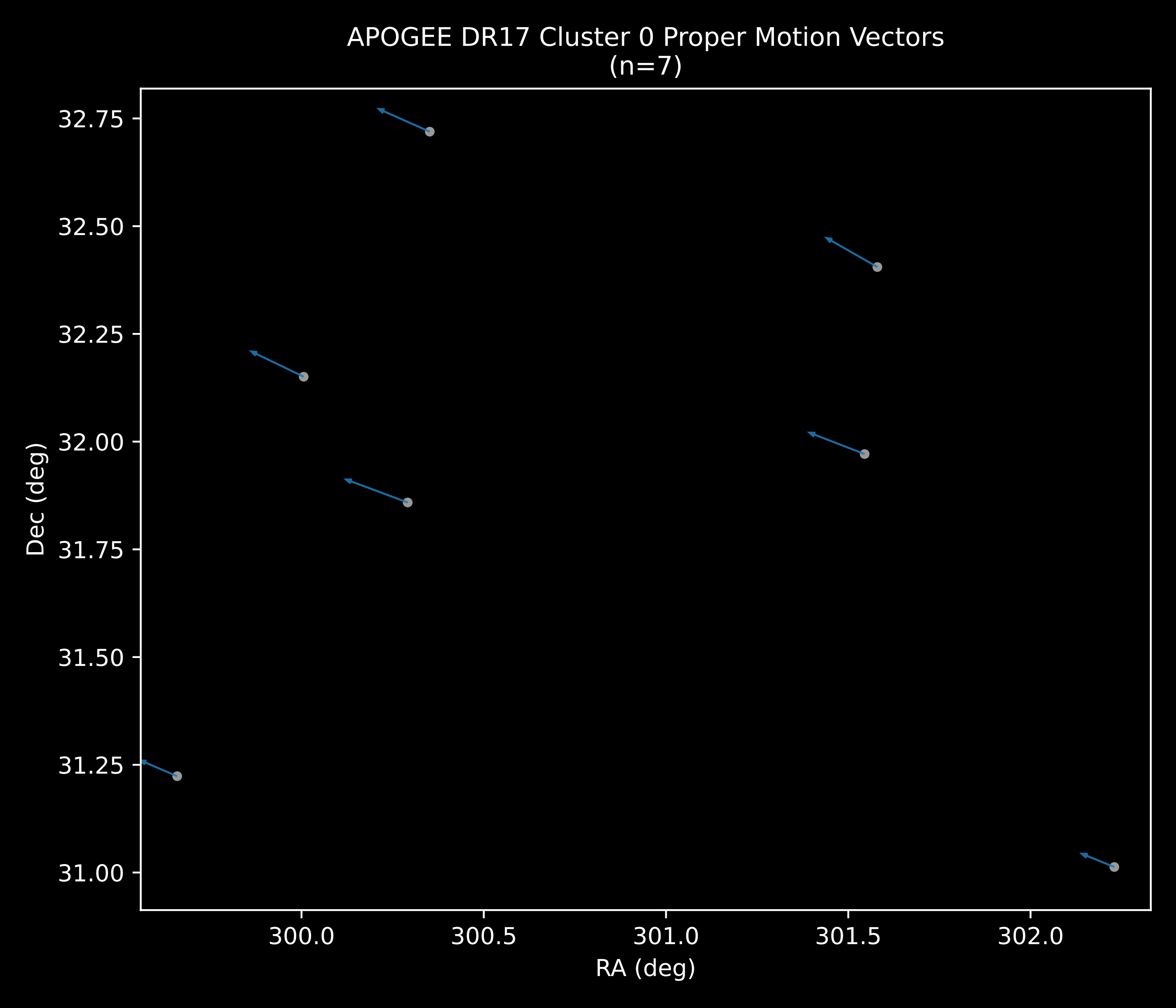


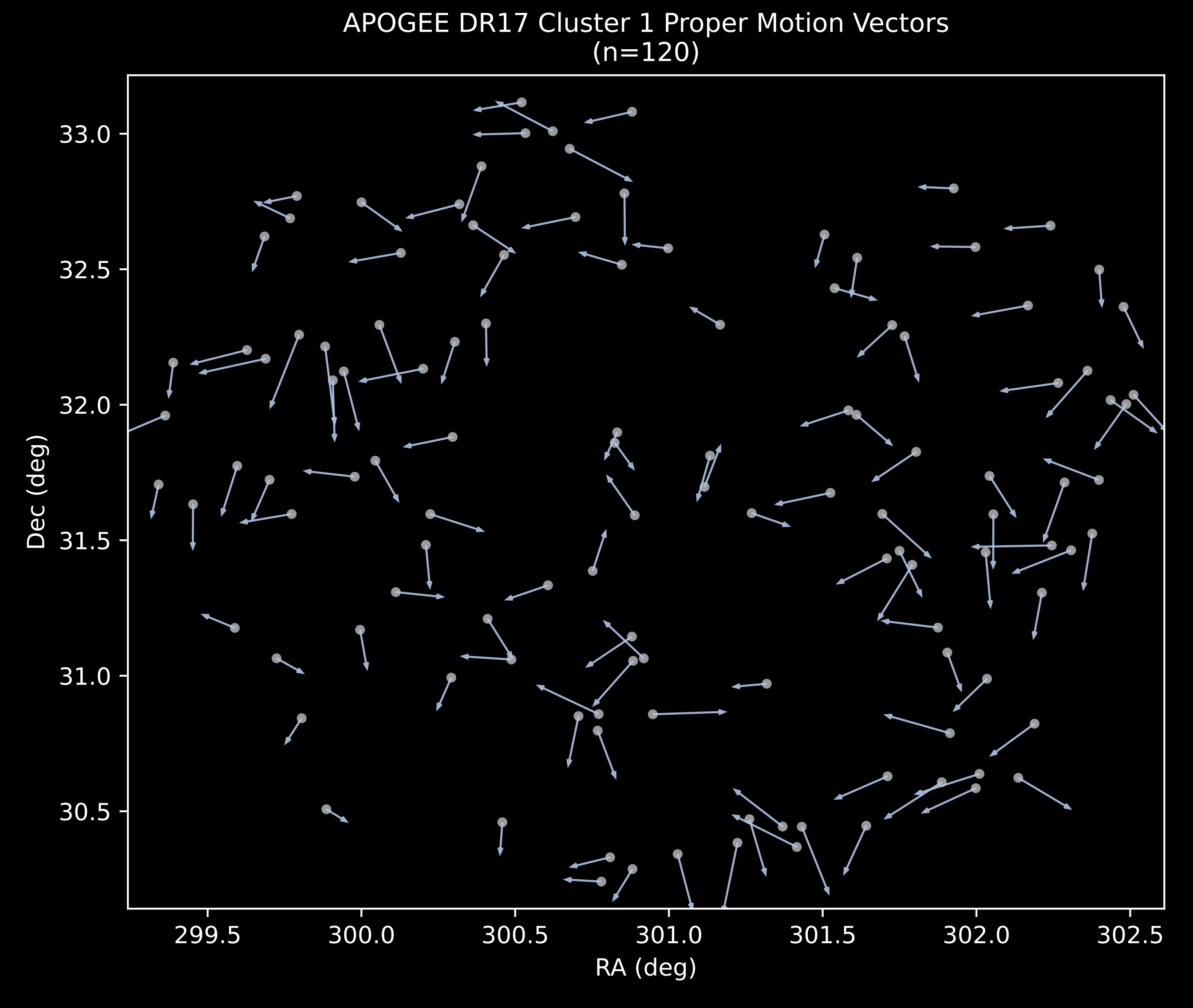
APOGEE DR17 Field: [068+82_MGA] Stellar Population Parameters (n= 175)
logg_gspphot
1.8
2.4
3.0
3.6
4.2
0.5
0.0
−0.5
−1.0
−1.5
−2.0
Metallicity ([M/H]) (gspphot)
11000
10000
9000
8000
7000
6000
5000
4000
Effective Temperature (T_{eff}) [K]


GAPOGEE DR17 Field [068+82_MGA] RUWE Distribution (n= 165)
RUWE = 1.4 Quality Cut
Star Count
RUWE (Renormalised Unit Weight Error)
0
10
20
30
40
50
60
70
1
2
3
4
5

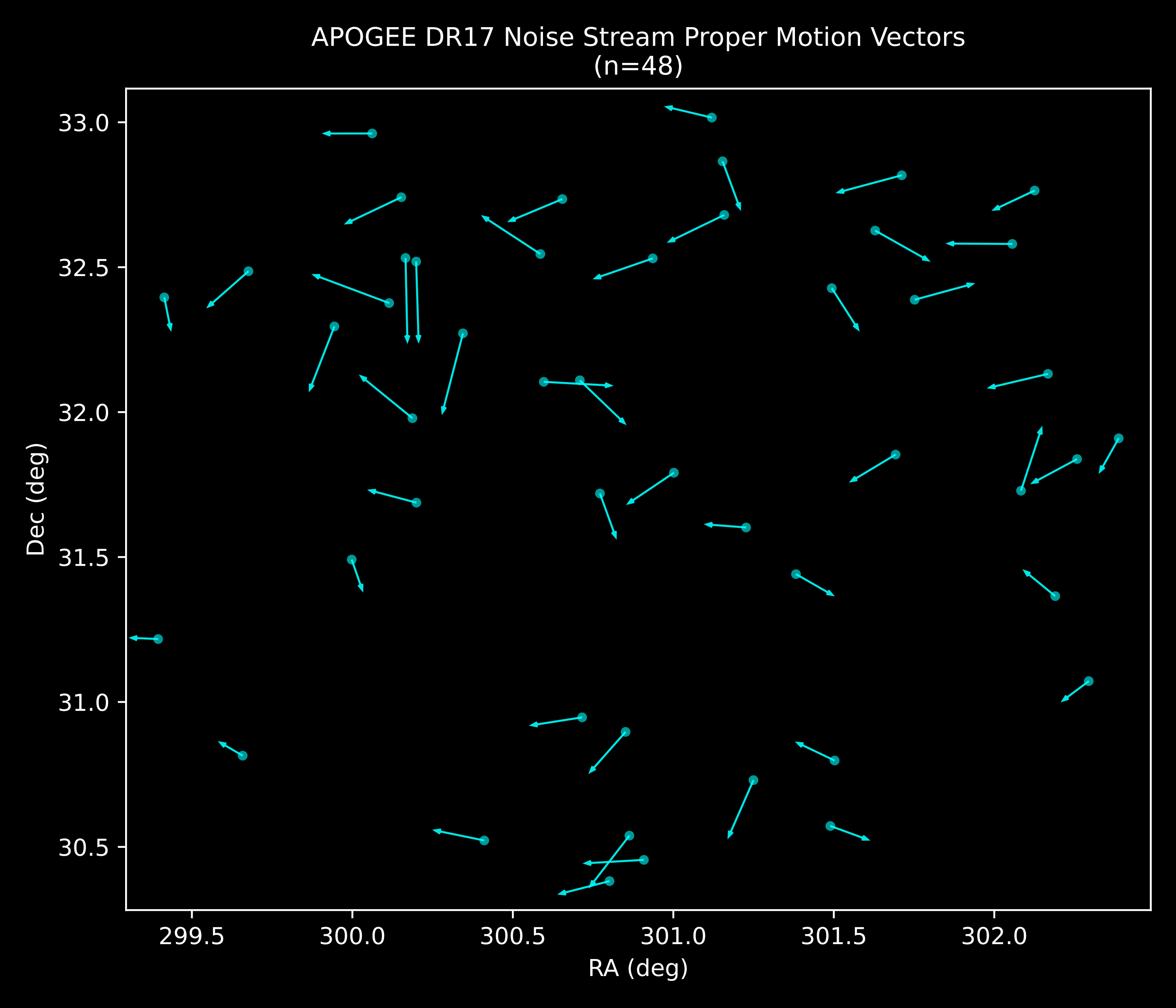

APOGEE DR17 Field: [068+82_MGA]
Proper Motion Vectors for Gaia Star Field (n= 175)
Stars
Dec (deg)
33.0
32.5
32.0
31.5
31.0
30.5
RA (deg)
299.5
300.0
300.5
301.0
301.5
302.0
302.5


APOGEE DR17 Field [068+82_MGA]
Proper Motion Vectors with HDBSCAN
(n=175)
HDBSCAN Clusters
Stars
Cluster 0 (n=7)
Cluster 1 (n=120)
Noise (n=48)
Dec (deg)
RA (deg)
33.0
32.5
32.0
31.5
31.0
30.5
299.5
300.0
300.5
301.0
301.5
302.0
302.5


APOGEE DR17 Cluster 0 Proper Motion Vectors
(n=7)
32.75
32.50
32.25
32.00
31.75
31.50
31.25
31.00
Dec (deg)
300.0
300.5
301.0
301.5
302.0
RA (deg)



APOGEE DR17 Cluster 1 Proper Motion Vectors
(n=120)
Dec (deg)
RA (deg)
33.0
32.5
32.0
31.5
31.0
30.5
299.5
300.0
300.5
301.0
301.5
302.0
302.5



APOGEE DR17 Noise Stream Proper Motion Vectors
(n=48)
33.0
32.5
32.0
31.5
31.0
30.5
Dec (deg)
299.5
300.0
300.5
301.0
301.5
302.0
RA (deg)